

JUNHAO ZHAO 赵俊豪

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👤 RESEARCH PROFILE

I am a Ph.D. student at **Xi'an Jiaotong-Liverpool University** (since September 2026), supervised by Dr. Lijie Yao. My research lies at the intersection of **visualization**, **human-computer interaction**, and **AI agents**. I am particularly interested in visualization in motion, interactive authoring tools, and reliable AI-assisted data communication. Previously, I studied artificial intelligence and 3D scene perception with Prof. Chenglu Wen at Xiamen University.

🎓 EDUCATION

Xi'an Jiaotong-Liverpool University, Suzhou, China Sep 2026 – Present

Ph.D. Student. Advisor: Dr. Lijie Yao.

Xiamen University, Xiamen, China Sep 2020 – Jun 2023

M.Sc. in Artificial Intelligence. Advisor: Prof. Chenglu Wen.

Nanjing Audit University, Nanjing, China Sep 2014 – Jun 2018

B.Sc. in Information Management and Information Systems.

🔬 RESEARCH EXPERIENCE

Research Assistant, Xi'an Jiaotong-Liverpool University Apr 2025 – Aug 2026

Supervisor: Dr. Lijie Yao

- Designed and implemented **SwimComposer**, an interactive authoring environment for coordinating data layers, views, transitions, and temporal composition in swimming-race videos; the work was accepted by IEEE TVCG for presentation at IEEE VIS 2026.
- Developed a multimodal data-preparation pipeline that aligns video tracking, commentary-derived insights, athlete information, and official records into visualization-ready data.
- Developed structured AI agents that produce executable and verifiable outputs for reliable visualization report generation, published at the IEEE VIS 2025 VISxGenAI workshop.

Weakly Supervised Learning for Dynamic 3D Scene Perception Jan 2022 – Jun 2023

NSFC General Program, Grant No. 62171393

- Designed weakly supervised learning strategies for sequential point clouds and evaluation methods for improving pseudo-label quality through multi-perspective temporal analysis.
- The work resulted in a first-authored IEEE TGRS paper and a Chinese national invention patent application.

Multi-platform Multimodal Point-cloud Data Processing May 2022 – Dec 2022

National Key R&D Program Young Scientists Project, Grant No. 2021YFF0700188

- Supported multimodal data preprocessing, heterogeneous-platform integration, and software development for scalable point-cloud management.

PUBLICATIONS

- [1] **Junhao Zhao**, Romain Vuillemot, Petra Isenberg, and Lijie Yao. “Authoring Narrative Visualization in Motion: Visual Storytelling in Swimming Videos.” *IEEE Transactions on Visualization and Computer Graphics*. Accepted; to be presented at **IEEE VIS 2026**.
- [2] **Junhao Zhao** and Lijie Yao. “Structured AI Agents for Reliable Visualization Report Generation.” *IEEE VIS 2025 Workshop on GenAI, Agents, and the Future of VIS (VISxGenAI)*.
- [3] **Junhao Zhao**, Weijie Huang, Hai Wu, Chenglu Wen, Bo Yang, Yulan Guo, and Cheng Wang. “SemanticFlow: Semantic Segmentation of Sequential LiDAR Point Clouds from Sparse Frame Annotations.” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1–11, 2023.

PATENT

Semantic Segmentation of Sequential Point Clouds from Sparse Labels Published Dec 2022

Chenglu Wen, Weijie Huang, **Junhao Zhao**, and Cheng Wang.
Chinese Patent Publication No. *CN115496899A*.

AWARD

Second Prize, Microsoft Edge Add-ons Innovation Challenge Sep 2022

Ranked 2nd among 148 teams. Independently developed *Joy-Reading*, a browser extension using MobileNet and TensorFlow.js to detect tables, formulas, and other key elements for more efficient document reading.

INDUSTRY EXPERIENCE

Software Product Manager, SND Group, Suzhou, China Jul 2023 – Jul 2024

Contributed to product design and development for smart agriculture and cultural-tourism platforms supporting enterprise digital transformation.

Data Engineer, Green Data Environmental Service Center Sep 2018 – Mar 2019

Suzhou, China. Collected, processed, and visualized pollution data from more than 20 cities for a large-scale environmental monitoring system.

TEACHING EXPERIENCE

ISA401 Human-AI Interaction Sep 2026 – Dec 2026

Teaching Assistant, Xi'an Jiaotong-Liverpool University.

SELECTED PRESENTATIONS

Presented at the IEEE VIS 2025 VISxGenAI workshop. Nov 2025
Attended the China 3D Vision Conference (China3DV). Apr 2023

LANGUAGES

Chinese (Mandarin): native · English: professional working proficiency